

Q8(a)(i)

Method 1: As the coil enters the field, there is an increase in magnetic flux linkage into the plane of the coil. By Faraday's Law, an e.m.f. and a current are induced in the coil. [1]

By Lenz's Law, the induced current flows in an anti-clockwise direction. [1]

This is to create an induced magnetic field pointing out of the plane of the coil to oppose the increase in flux linkage. [1]

Method 2: As the coil moves to the right, there is an increase in magnetic flux linkage into the plane of the coil. By Faraday's Law, an e.m.f. and a current are induced in the coil. [1]

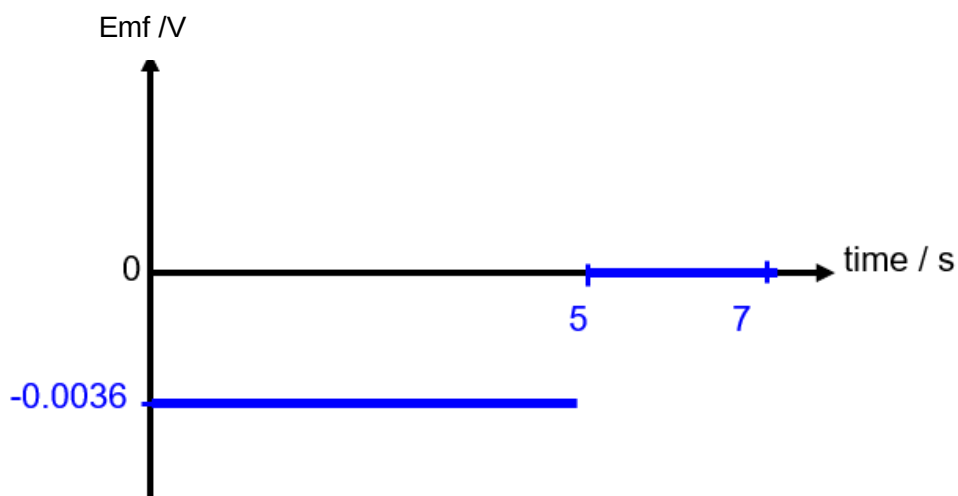
Fleming's Left Hand Rule predicts an induced current flowing anticlockwise in the coil. [1]

This is so that an induced magnetic force will be directed leftward on arm QR to oppose the motion of the coil into the magnetic field. [1]

Q8(a)(ii)

$$\phi_{\max} = BA = 1.5 \times 0.120 \times 0.100 = \mathbf{0.018 \text{ Wb}} \quad [1]$$

Q8(a)(iii)



Horizontal line with e.m.f. = -0.0036 V, starting from 0 to 5 s [1]

Horizontal line with e.m.f. = 0 V, starting from 5 to 7 s [1]

[No mark awarded without correct label.]

Q8(b)(i)

The magnetic field is directed out of the plane of the paper. [1]

Q8(b)(ii)

The protons experience a magnetic force. [1]

As the magnetic force is always perpendicular to the direction of travel of the proton (or the magnetic force is always directed towards a point – the centre of a circle), the proton moves in a circular motion. [1]

Q8(b)(iii)

The magnetic force provides centripetal force for circular motion of the protons. [1]

$$Bev = m \frac{v^2}{r} \quad [1]$$

$$r = \frac{mv}{Be}$$

Q8(b)(iv)

$$\text{Time spent in one semi-circle} = \frac{0.5 \times \text{circumference}}{\text{speed}} \quad [1]$$

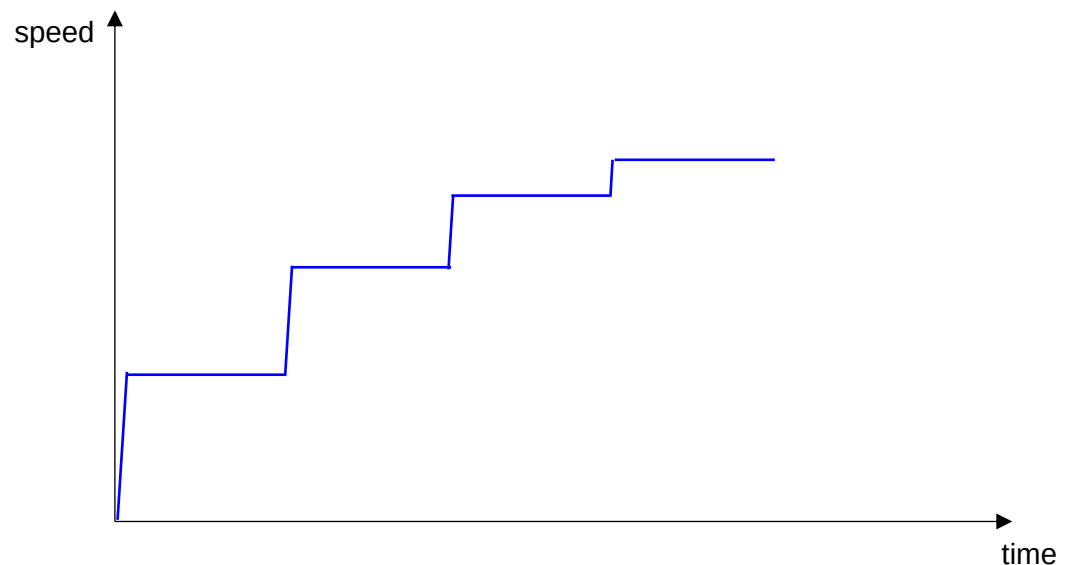
$$\text{OR: time spent in one semi-circle} = \frac{\text{period}}{2}$$

$$t = \frac{\pi r}{v} = \frac{\pi}{v} \left(\frac{mv}{Be} \right) = \frac{\pi m}{Be} \quad [1]$$

Q8(b)(v)

The time of travel in the dee, $t = \frac{\pi m}{Be}$, is independent of the speed of the proton. [1]

Q8(b)(vi)



4 steps (as asked in question) [1]

Duration of each step constant (as discussed in (b)(v)) [1]

Step-up progressively smaller and time between steps is very short. [1]

Q8(c)(ii)

The output energy of the particle is the kinetic energy when the particle moves in the biggest semi-circle available at radius R .

$$R = \frac{mv}{Be}$$

$$v^2 = \frac{B^2 e^2 R^2}{m^2} \quad [3]$$

$$E = \frac{1}{2}mv^2 = \frac{B^2 e^2 R^2}{2m}$$

[1st mark for the statement or equivalent. The next 2 marks for the correct manipulation to reach the expression.]

***** END *****